

Mandibuloacral Dysplasia Type B (MAD-B): Comprehensive Disease Characteristics Report

Disease: Mandibuloacral Dysplasia Type B (MAD-B / MADB) **MONDO ID:** MONDO:0012074 · **OMIM:** #608612 · **Causal gene:** *ZMPSTE24* (FACE1), OMIM 606480, HGNC:12877, chromosome 1p34.2 **Category:** Mendelian, autosomal recessive progeroid laminopathy

Summary

Mandibuloacral dysplasia type B (MAD-B) is an ultra-rare, autosomal recessive progeroid laminopathy caused by biallelic **hypomorphic** (residual-activity) mutations in **ZMPSTE24 (FACE1)**, the zinc metalloprotease that performs the final endoproteolytic maturation of prelamin A into mature lamin A. When ZMPSTE24 activity is reduced, toxic **farnesylated, carboxymethylated prelamin A** accumulates at the inner nuclear membrane, disrupting nuclear lamina architecture and triggering a cascade of downstream pathology: defective DNA-damage repair, epigenetic/chromatin dysregulation (H4K16 hypoacetylation, aberrant H3K9me3), AKT-mTOR pathway activation, oxidative/mitochondrial stress, and premature cellular senescence. These converge on the characteristic multisystem clinical picture of growth failure, craniofacial and skeletal dysplasia (mandibular and clavicular hypoplasia, delayed cranial suture closure, acro-osteolysis of distal phalanges), **generalized (type B) lipodystrophy** with insulin resistance and metabolic complications, and progeroid skin/hair changes (mottled atrophic skin, sparse brittle hair, hypoplastic nails).

A central organizing principle of this disease is a **gene-dosage/enzyme-activity severity gradient** shared with its allelic disorder restrictive dermopathy (RD): complete ZMPSTE24 loss-of-function (biallelic null) causes neonatally lethal RD, whereas variants that preserve residual prelamin A-processing activity cause the milder, survivable MAD-B. This genotype-phenotype correlation is quantitatively supported across case series and is biologically mirrored in *Zmpste24*-deficient mouse models, which recapitulate the bone, muscle, lipodystrophy,

immune, and epigenetic phenotypes and have supported mechanism-targeted therapeutic proof-of-concept (isoprenylcysteine methylation inhibition, RCE1/ICMT targeting, SUV39H1 depletion, RANKL inhibition).

There is **no approved disease-modifying therapy** for MAD-B. Management is multidisciplinary and symptomatic, targeting the disease components: metabolic control of the lipodystrophy/diabetes/hypertriglyceridemia (including leptin replacement with **metreleptin**, the only approved therapy for generalized lipodystrophy), bone/orthopedic care, and dermatologic and dental surveillance. Prenylation-targeted drugs borrowed from the Hutchinson–Gilford progeria (HGPS) paradigm — notably the farnesyltransferase inhibitor **lonafarnib**, which rescues nuclear morphology in MAD-B patient fibroblasts — represent the most mechanistically rational experimental strategy.

Evidence-source key: **[H]** human clinical · **[M]** model organism · **[V]** in vitro · **[C]** computational/database. Primary citations are given as PMIDs; database-level identifiers are aggregated resource knowledge.

1. Disease Information

Overview. MAD-B is a rare autosomal recessive progeroid disorder of the mandibuloacral dysplasia (MAD) spectrum, defined molecularly by mutations in *ZMPSTE24* and clinically by the combination of mandibular and clavicular hypoplasia, acro-osteolysis, delayed cranial suture closure, generalized lipodystrophy, and progeroid skin features. It belongs to the **laminopathies** — disorders of the nuclear lamina — and specifically to the subgroup caused by defective post-translational processing of prelamin A.

Key identifiers.

Resource	Identifier
MONDO	MONDO:0012074
OMIM (phenotype)	#608612 (Mandibuloacral dysplasia with type B lipodystrophy; MADB)
Causal gene	<i>ZMPSTE24</i> / <i>FACE1</i> (OMIM 606480, HGNC:12877, 1p34.2)
Disease class	Autosomal recessive progeroid laminopathy

Synonyms / alternative names. Mandibuloacral dysplasia with type B lipodystrophy; MADB; MAD type B; MAD-B; ZMPSTE24-related mandibuloacral dysplasia; mandibuloacral dysplasia with generalized lipodystrophy.

Information source. The knowledge base here derives from **aggregated resources plus individual patient case reports and small case series** — MAD-B is ultra-rare, and the strongest clinical dataset is an aggregate of 20 genetically confirmed patients, anchored by an 8-patient Suriname founder cohort [H] (PMID: 31856865).

2. Etiology

Primary cause — genetic. MAD-B is caused by **biallelic (homozygous or compound heterozygous) mutations in ZMPSTE24**, reducing zinc-metalloprotease activity and impairing prelamin A processing. This is a Mendelian, monogenic disease; it is not infectious, and there is no primary environmental cause.

"MADB is caused by reduced activity of the enzyme zinc metalloprotease ZMPSTE24 resulting from compound heterozygous or homozygous mutations in ZMPSTE24." — [H] PMID: 31856865

"Mutation in ZMPSTE24 gene, encoding a major metalloprotease, leads to defective prelamin A processing and causes type B mandibuloacral dysplasia, as well as the lethal neonatal restrictive dermopathy syndrome." — [H] PMID: 21267004

Genetic risk factors. The causal variants themselves are the risk determinant. Severity depends on the **residual enzyme activity** conferred by the specific genotype (Finding F002). **Consanguinity and founder effects** increase recurrence risk — the Suriname cohort (n=8) is homozygous for a single founder allele, c.1196A>G p.(Tyr399Cys).

Environmental / iatrogenic modifiers. Although MAD-B is genetic, the same enzymatic pathway can be inhibited pharmacologically: **HIV protease inhibitors** (notably lopinavir) inhibit ZMPSTE24 and phenocopy prelamin A pathology, causing acquired lipodystrophy (Finding F009). This is an acquired/iatrogenic analogue rather than a cause of Mendelian MAD-B, but it establishes a key gene–environment convergence on ZMPSTE24.

Protective factors. No established genetic or environmental protective alleles are documented for MAD-B. Mechanistically, interventions that **reduce prelamins A prenylation/methylation** (e.g., ICMT or RCE1 loss, FTIs) are protective in models — a therapeutic inference, not a naturally occurring protective factor.

3. Phenotypes

The phenotype is a multisystem progeroid syndrome. Major clinical criteria were defined across all 20 genetically confirmed patients, each present in **85–100%** of affected individuals [H] ([PMID: 31856865](#)).

"Major criteria were found to be: short stature, clavicular hypoplasia, delayed closure of cranial sutures, high palate, mandibular hypoplasia, dental crowding, acro-osteolysis of the distal phalanges, hypoplastic nails, brittle and/or sparse hair, mottled pigmen[tation]..." — [PMID: 31856865](#)

"MADB is characterized by brittle hair, mottled, atrophic skin, generalized lipodystrophy, insulin resistance, metabolic complications and skeletal features like stunted growth, mandibular and clavicular hypoplasia and acro-osteolysis of the distal phalanges." — [PMID: 31856865](#)

Phenotype	Type	Suggested HPO term	Frequency	Onset / course
Short stature / growth failure	Physical	HP:0004322	High (major; 85–100%)	Childhood; progressive
Mandibular hypoplasia	Clinical sign	HP:0000347	High (major)	Childhood
Clavicular hypoplasia	Clinical sign	HP:0006660	High (major)	Childhood
Delayed cranial suture closure	Clinical sign	HP:0000270	High (major)	Childhood
High/narrow palate	Clinical sign	HP:0000218	High (major)	Congenital/ childhood
Dental crowding	Clinical sign	HP:0000678	High (major)	Childhood
Acro-osteolysis of distal phalanges	Radiographic	HP:0009771	High (major)	Childhood; progressive
Hypoplastic nails	Physical	HP:0001792	High (major)	Childhood
Brittle / sparse hair	Physical	HP:0002213 / HP:0008070	High (major)	Childhood
Mottled pigmentation / atrophic skin	Physical	HP:0001000 / HP:0000963	High (major)	Childhood; progressive
Generalized (type B) lipodystrophy	Physical/ metabolic	HP:0009125	Core feature	Childhood; progressive
Insulin resistance / diabetes mellitus	Laboratory	HP:0000855 / HP:0000857	Common	Childhood– adolescence
Hypertriglyceridemia	Laboratory	HP:0002155	Common	Childhood– adolescence
Congenital myopathy (reported)	Clinical sign	HP:0003198	Variable/ rare	Homozygous missense case (PMID: 21267004)

Severity / progression. Features are generally **progressive and chronic** (bone resorption, lipodystrophy, skin atrophy worsen over time), but the overall course is slowly progressive and survivable into adulthood (contrast lethal RD — Section 8). **Expressivity is variable**, even within the same genotype.

Quality-of-life impact. Short stature, craniofacial/orthopedic abnormalities, and skin/hair changes affect appearance, dentition, and mobility; metabolic complications (diabetes, hypertriglyceridemia) impose chronic disease burden. Disease-specific validated QoL instruments for MAD-B are not available; impact is inferred from the lipodystrophy and progeroid literature.

4. Genetic / Molecular Information

Causal gene. *ZMPSTE24* (also *FACE1*; zinc metallopeptidase STE24), encoding a membrane-bound zinc metalloprotease of the inner nuclear/ER membrane that cleaves farnesylated prelamin A. Gene OMIM 606480; phenotype OMIM #608612.

Variant spectrum (Finding F007). MAD-B variants span **missense, frameshift, and splice-site** classes, unified by preservation of some residual enzyme activity:

Variant (cDNA)	Protein	Type	Note	Source
c.1196A>G	p.(Tyr399Cys)	Missense	Suriname homozygous founder allele (n=8)	PMID: 31856865
c.281T>C	p.(Leu94Pro)	Missense	Reported MAD-B	literature spectrum
c.794A>G	p.(Asn265Ser)	Missense	Compound het with frameshift	PMID: 30919593
c.1085dup	p.(Leu362Phefs*19)	Frameshift	Compound het (Chile case)	PMID: 30919593
c.1077dupT	p.(Leu362fsX18)	Frameshift	Mosaic UPD chr1; intermediate phenotype	PMID: 29341437
c.28_29insA	p.(Leu10Tyrfs*37)	N-terminal frameshift	Rescued by downstream alternative start codon	PMID: 35597529
c.378+1G>A	—	Splice-site	Homozygous (Chinese MAD case report)	PMID: 38544690

A notable molecular phenomenon is **alternative translation initiation rescue**: N-terminal frameshifts predicted to be null instead use a downstream in-frame start codon, yielding a hypomorphic (residual-activity) protein that produces survivable MAD-B rather than lethal RD.

"the one base pair insertion creates a novel downstream in-frame start codon, which supposedly serves as an alternative translation initiation site (TIS)" — [H/V] PMID: 35597529

"mutations preserving residual enzymatic activity of the ZMPSTE24 protein lead to the milder mandibuloacral dysplasia with type B lipodystrophy (MADB) phenotype" — [H/V] PMID: 37270786

Variant classification. Reported MAD-B alleles are classified **pathogenic/likely pathogenic** (ACMG/AMP), supported by segregation, recessive genotype, functional reduction in prelamin A processing, and phenotype concordance. **Allele frequencies** in population databases (gnomAD) are very low/absent, consistent with recessive disease.

Functional consequence. Loss of function of ZMPSTE24 protease activity → toxic accumulation of **farnesylated prelamin A** (a gain-of-toxic-function at the substrate level). Severity scales inversely with residual activity (Finding F002).

Origin. **Germline**, biallelic. One reported patient had an unusual **mosaic uniparental isodisomy of chromosome 1** unmasking a ZMPSTE24 variant ([PMID: 29341437](#)).

Modifier genes / epigenetics. In models, modifiers of the prelamin A pathway alter severity: *SUV39H1* (H3K9 methyltransferase) depletion improves DNA repair and lifespan; *ICMT* and *RCE1* (prenyl-processing enzymes) modulate prelamin A toxicity. Epigenetic changes downstream of ZMPSTE24 loss include **H4K16 hypoacetylation** and altered H3K9me3 (Section 6).

5. Environmental Information

MAD-B is a **monogenic genetic disease**; no environmental, lifestyle, or infectious agent causes it. However, the **ZMPSTE24 enzyme is a documented pharmacologic target**: HIV protease inhibitors inhibit ZMPSTE24 and phenocopy prelamin A disease (acquired lipodystrophy, accelerated cellular senescence), establishing an environmental/iatrogenic analogue of the genetic defect (Finding F009).

"lopinavir accelerated cellular senescence by inhibiting Zmpste24 and interfering nuclear membrane stability, which leads to decreased binding between nuclear membrane-binding protein Usp7 and Mdm2 and activates Usp7/Mdm2/p53 pathway" — [M/H] [PMID: 40461512](#)

No infectious agents are implicated.

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Biallelic hypomorphic ZMPSTE24 mutation** reduces zinc-metalloprotease activity (*demonstrated*).

2. This **results in** defective endoproteolytic maturation of prelamin A → lamin A (*demonstrated*).
3. Which **leads to** accumulation of **farnesylated, carboxymethylated prelamin A** anchored at the inner nuclear membrane (*demonstrated*).
4. Accumulated prelamin A **disrupts nuclear lamina architecture / nuclear morphology** (misshapen nuclei) (*demonstrated in patient fibroblasts; rescued by FTI*).
5. Lamina disruption **branches** into several downstream effects:
6. **5a. Impaired DNA-damage repair** via **H4K16 hypoacetylation** (reduced Mof association with nuclear matrix) and aberrant **H3K9me3** (SUV39H1-dependent) → genomic instability (*demonstrated in mouse*).
7. **5b. Prelamin A-dependent activation of AKT–mTOR signaling** (prelamin A–AKT interaction) → altered proliferation/senescence balance (*demonstrated*).
8. **5c. Mitochondrial dysfunction and oxidative stress**, prominent in adipose tissue (*demonstrated proteomically*).
9. These **converge on premature cellular senescence** and tissue-specific stem/progenitor depletion (*demonstrated*).
10. **Senescence + progenitor depletion** in skeletal stem/progenitor cells, adipocytes, osteocytes, and other lineages **results in** the clinical phenotype: bone dysplasia/acro-osteolysis, generalized lipodystrophy with insulin resistance, progeroid skin/hair, and growth failure (*inferred from model + human concordance*).

Detailed mechanism

Molecular pathways. The central lesion is in the **prelamin A post-translational processing pathway** (farnesylation → –AAX cleavage → carboxymethylation → ZMPSTE24-mediated 15-residue upstream cleavage to mature lamin A). Downstream signaling nodes include **AKT–mTOR** (activated by prelamin A), the **Usp7/Mdm2/p53** senescence axis (shown pharmacologically with lopinavir), and **NF-κB** (B-lymphopoiesis in marrow).

"triggered prelamin A-dependent activation of AKT-mammalian target of rapamycin (mTOR) signaling, which abolished the premature senescence of Zmpste24-deficient fibroblasts." — [M/V] PMID: 23686339

Cellular processes. Premature **cellular senescence** (GO:0090398), **cell-cycle dysregulation**, **DNA-damage response** (GO:0006974), and **stem/progenitor cell depletion** are core. scRNA-seq of conditional *Zmpste24* deletion shows loss of skeletal stem/progenitor cell populations drives bone loss [M] (PMID: 37407584).

Protein dysfunction. Loss of ZMPSTE24 catalytic function → persistent farnesyl/methyl modification of prelamin A, which anchors it to the nuclear envelope as a toxic species (gain-of-toxic-function at the substrate).

Epigenetic changes. **H4K16 hypoacetylation** (reduced Mof HAT at nuclear matrix) and **SUV39H1-dependent H3K9me3** dysregulation link lamina disruption to defective DNA repair.

"histone H4 was hypoacetylated at a lysine 16 residue (H4K16), and this defect was attributed to the reduced association of a histone acetyltransferase, Mof, to the nuclear matrix" — [M] PMID: 21746928

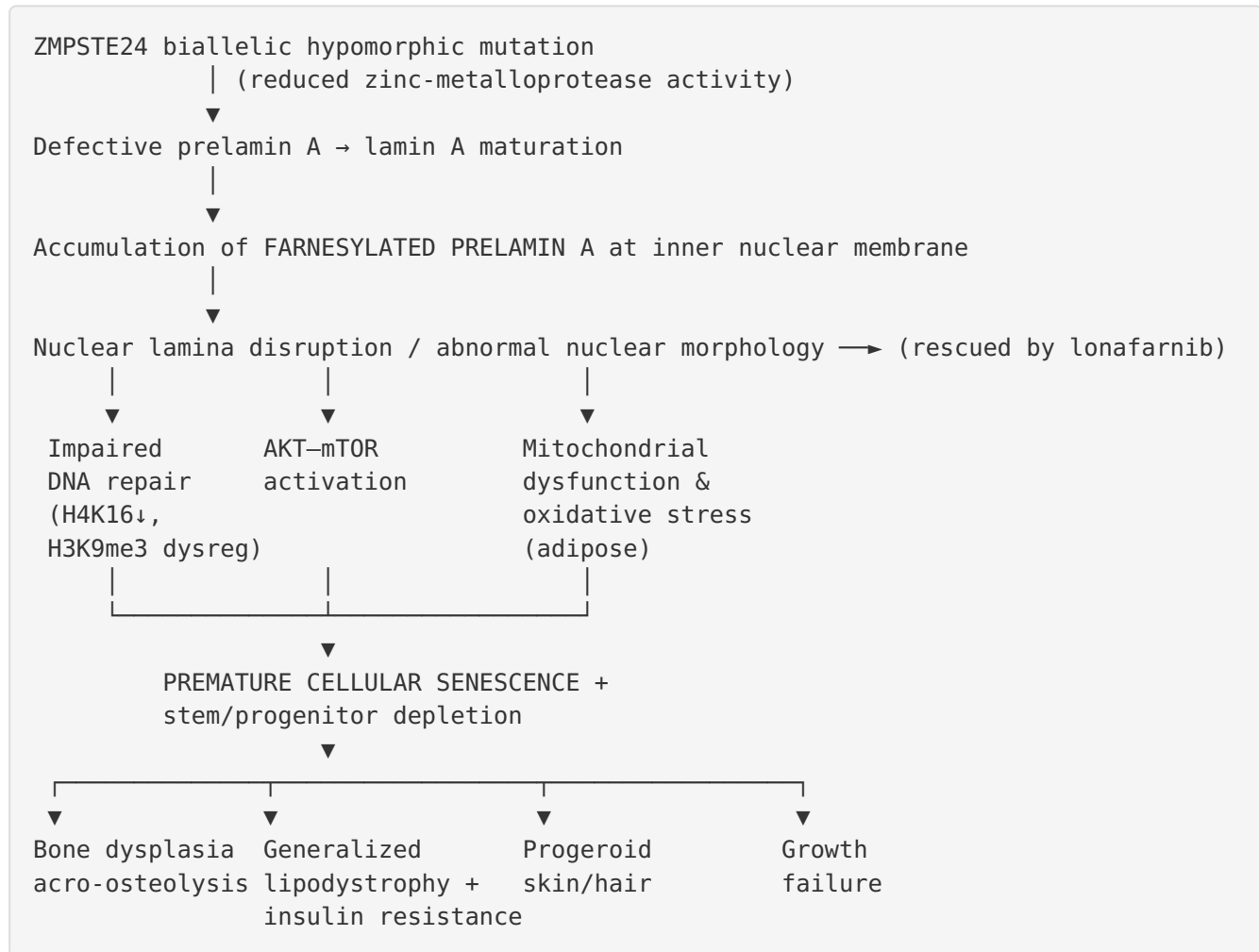
Metabolic / mitochondrial. Adipose-tissue proteomics in *Zmpste24*^{-/-} mice reveal major mitochondrial changes and vimentin processing alterations, tying defective prelamin A processing to lipotrophy and oxidative damage.

"support the relationship between defective prelamin A processing and mitochondrial dysfunction and highlight the relevance of oxidative damage in lipotrophy and aging" — [M] PMID: 21828285

Immune involvement. Prelamin A accumulation compromises **NF-κB-regulated B-lymphopoiesis** via the bone-marrow microenvironment in the progeria mouse model [M] (PMID: 24764515).

Ontology suggestions. Biological processes: GO:0006508 (proteolysis), GO:0006974 (DNA damage response), GO:0090398 (cellular senescence), GO:0006629 (lipid metabolic process). Cell types: CL:0000136 (adipocyte), CL:0000062 (osteoblast), CL:0000137 (osteocyte), CL:0000057 (fibroblast), skeletal stem/progenitor cell. Cellular components: GO:0005637 (nuclear inner membrane), GO:0005635 (nuclear envelope), GO:0005739 (mitochondrion). Chemicals: CHEBI:24017 (farnesyl group); farnesyltransferase inhibitor class.

Mechanistic schematic



7. Anatomical Structures Affected

- **Skeletal system (primary).** Mandible (UBERON:0001684), clavicle (UBERON:0001105), cranial sutures (UBERON:0007842), distal phalanges (acro-osteolysis; UBERON:0004248), skull/palate. Bone loss is driven by skeletal stem/progenitor cell senescence and osteocyte-RANKL signaling.
- **Adipose tissue (primary).** Generalized loss of subcutaneous and visceral adipose tissue (UBERON:0001013); mitochondrial dysfunction in adipocytes.
- **Skin and appendages.** Skin (UBERON:0002097) — mottled, atrophic; hair (brittle/sparse); nails (hypoplastic).

- **Endocrine/metabolic (secondary).** Insulin resistance, diabetes mellitus, hypertriglyceridemia — consequences of lipodystrophy and leptin deficiency.
- **Muscle.** Congenital myopathy reported in a homozygous missense case ([PMID: 21267004](#)); muscle wasting prominent in *Zmpste24*^{-/-} mice.
- **Hematopoietic/immune (model-level).** Impaired B-lymphopoiesis in the marrow microenvironment (mouse).
- **Subcellular.** Nuclear envelope / inner nuclear membrane (GO:0005637), nuclear lamina, mitochondria (GO:0005739).
- **Lateralization.** Skeletal features are generally **bilateral/symmetric**.

8. Temporal Development

Onset. **Postnatal / childhood** onset; chronic, insidious course (Finding F011). This contrasts sharply with the allelic **restrictive dermopathy (RD)**, the ZMPSTE24-null disorder with congenital taut translucent skin, joint contractures, and **death usually within the first month of life**.

"death usually occurring within the first month of life" (lethal neonatal RD) — **[H]** [PMID: 41877632](#)

Progression. Slowly progressive and chronic/lifelong; skeletal resorption, lipodystrophy, and skin atrophy worsen over time. A **30-year longitudinal clinical survey** of a ZMPSTE24-mutant progeroid patient documents survival into and through adulthood.

"we report a 30-year longitudinal clinical survey of a patient" — **[H]** [PMID: 21267004](#)

Severity continuum. Position on the RD ↔ MAD-B continuum is set by **residual ZMPSTE24 activity / prelamin A burden** (Finding F002). An "intermediate form" genocopying severe progeria has been described ([PMID: 29341437](#)).

Critical window. Mechanism-targeted interventions in models work best when started early (e.g., *Rce1* knockout at postnatal week 4–5 doubled median survival), suggesting an early-childhood window of opportunity.

9. Inheritance and Population

- **Inheritance. Autosomal recessive** (biallelic ZMPSTE24).
- **Penetrance.** Effectively complete for biallelic pathogenic genotypes; **expressivity is variable** across (and within) genotypes.
- **Prevalence/incidence.** Ultra-rare; precise prevalence not established. Only ~20 genetically confirmed patients form the aggregate clinical dataset; reported as isolated cases and small series worldwide.
- **Founder effects / consanguinity.** A **homozygous founder mutation (c.1196A>G, p.Tyr399Cys)** underlies the 8-patient **Suriname** cohort — a clear founder effect. Consanguinity increases recurrence risk; several reported cases arise from consanguineous or isolated-population backgrounds.
- **Geographic distribution.** Reported globally (Suriname, Chile, China, and others); no single endemic region beyond founder clusters.
- **Sex ratio.** Autosomal recessive — expected **1:1 male:female**.
- **Carrier frequency.** Not formally established; expected very low given rarity.

10. Diagnostics

Clinical/imaging. Diagnosis is suspected from the characteristic constellation: short stature, mandibular/clavicular hypoplasia, delayed cranial suture closure, high palate, dental crowding, **radiographic acro-osteolysis of distal phalanges**, hypoplastic nails, sparse brittle hair, mottled atrophic skin, and generalized lipodystrophy. Skeletal survey/radiography demonstrates clavicular hypoplasia and acro-osteolysis. **Metabolic labs** show insulin resistance, hyperglycemia/diabetes, and hypertriglyceridemia.

Cellular/functional. Skin fibroblasts show **abnormal nuclear morphology** (misshapen nuclei) from prelamin A accumulation — a supportive in vitro finding reversible with FTI (lonafarnib).

Genetic testing (confirmatory, gold standard). - **Single-gene sequencing of ZMPSTE24** confirms the diagnosis. - **Gene panels** (lipodystrophy / laminopathy / progeroid panels including *LMNA*, *ZMPSTE24*, *POLD1*, *MTX2*) and **whole-exome sequencing (WES)** are

effective when phenotypes overlap other MAD/progeroid syndromes. - **WGS/WES** also detect unusual mechanisms (e.g., mosaic uniparental isodisomy of chromosome 1 unmasking a ZMPSTE24 allele — PMID: 29341437). - Functional confirmation: reduced prelamin A processing / prelamin A accumulation on immunoblot.

Clinical diagnostic criteria. Formal **major/minor criteria and management guidelines** were proposed from the 20-patient aggregate (PMID: 31856865; Finding F003).

Differential diagnosis (Finding F006).

Disorder	Gene	Distinguishing feature
MAD type A	LMNA	Partial (type A) lipodystrophy
MAD type B	ZMPSTE24	Generalized (type B) lipodystrophy
Restrictive dermopathy	ZMPSTE24 (null)	Neonatal lethal; taut translucent skin, contractures
Hutchinson–Gilford progeria	LMNA (progerin)	Severe vascular disease; distinct skeletal pattern
MDPL	POLD1	Deafness, male hypogonadism
MDPS	MTX2 (null)	Arterial calcification, renal glomerulosclerosis, hypertension
MDP syndrome	unknown	Sensorineural deafness; no clavicular hypoplasia/acro-osteolysis (PMID: 20631028)
Hallermann–Streiff syndrome	unknown (not a laminopathy)	Congenital cataracts; no ZMPSTE24/LMNA pathogenic variants (PMID: 22570643)

"The molecular defects associated with MAD are mutations in LMNA or ZMPSTE24 (FACE1) gene, causing type A or type B MAD, respectively." — [H] PMID: 29208544

"We report five homozygous null mutations in MTX2, encoding Metaxin-2 (MTX2), an outer mitochondrial membrane protein, in patients presenting with a severe laminopathy-like mandibuloacral dysplasia" — [H] PMID: 32917887

Screening. Carrier screening / cascade testing in affected families (especially founder populations); prenatal / preimplantation genetic diagnosis is feasible once familial variants are known.

11. Outcome / Prognosis

- **Survival.** Unlike lethal neonatal RD, MAD-B is **survivable into adulthood**; a 30-year longitudinal survey documents long-term survival ([PMID: 21267004](#)). Prognosis is set by residual enzyme activity.
 - **Morbidity.** Substantial chronic morbidity from skeletal dysplasia (acro-osteolysis, fractures, deformity), lipodystrophy-driven **metabolic disease** (diabetes, severe hypertriglyceridemia with pancreatitis risk, hepatic steatosis), and progeroid skin/dental problems.
 - **Complications.** Metabolic complications of generalized lipodystrophy; orthopedic complications; potential cardiac repolarization abnormalities reported in prelamin A disorders (progeroid models/HGPS) warrant surveillance.
 - **Prognostic factors.** Degree of residual ZMPSTE24 activity / prelamin A accumulation (molecular prognostic axis); severity of metabolic complications.
 - **Recovery.** No cure; recovery is not expected, but metabolic management substantially modifies morbidity.
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12. Treatment

No disease-modifying therapy is approved for MAD-B; management is multidisciplinary and symptomatic (Finding F010). Management guidelines accompany the diagnostic criteria ([PMID: 31856865](#)).

Metabolic / lipodystrophy management. - **Metreleptin** (recombinant leptin; NCIT: leptin replacement therapy / metreleptin) — the **only approved therapy for generalized lipodystrophy** and mechanistically applicable to MAD-B's leptin-deficient generalized lipodystrophy. In generalized lipodystrophy cohorts it produced significant reductions in HbA1c and triglycerides.

"Significant short-term median reductions from baseline were observed for HbA1c (- 1.4%; n = 26), fasting plasma glucose (- 0.5 mmol/L; n = 21); fasting triglycerides (TG; -1.5 mmol/L; n = 26)" — [H] PMID: 42507316

- Standard metabolic agents (insulin sensitizers, lipid-lowering agents), diet, and exercise for diabetes/hypertriglyceridemia.

Mechanism-targeted / experimental (prenylation-directed). - Lona-farnib (farnesyltransferase inhibitor; NCIT: farnesyltransferase inhibitor) **improves abnormal nuclear morphology in ZMPSTE24-deficient MAD-B patient fibroblasts**, providing direct in vitro rationale.

"Several related progeroid disorders are caused by defective post-translational processing of prelamin A, the precursor of the nuclear scaffold protein lamin A..." — [V] PMID: 38050983

- The HGPS **triple-therapy** paradigm (lona-farnib + pravastatin + zoledronic acid) improved bone mineral density and weight gain in trial, supporting translation to prelamin A disorders.

"Secondary improvements included increased areal ($P=0.001$) and volumetric ($P<0.001-0.006$) bone mineral density and 1.5- to 1.8-fold increases in radial bone structure ($P<0.001$)." — [H] PMID: 27400896

- Model-level strategies with translational promise: **ICMT inhibition** (delays senescence), **RCE1 targeting** (doubled median survival in mice), **SUV39H1 depletion** (~60% lifespan extension in mice), **RANKL inhibition** (antiresorptive; improves bone, muscle, lifespan in progeroid mice), and **mTOR modulation**.

Supportive / surgical. Orthopedic/dental surveillance and intervention; physical therapy; dermatologic care; growth and nutritional management.

Pharmacovigilance. Avoid HIV protease inhibitors where alternatives exist, since they inhibit ZMPSTE24 and could worsen prelamin A pathology (Finding F009).

13. Prevention

- **Primary prevention.** Not possible for a Mendelian disease; prevention centers on **genetic counseling** and reproductive options.
 - **Genetic counseling / carrier & cascade testing.** Autosomal recessive recurrence risk of 25% for carrier couples; cascade testing in founder populations (e.g., Suriname). **Prenatal diagnosis and preimplantation genetic testing** available once familial variants are known.
 - **Secondary prevention.** Early detection and management of metabolic complications (diabetes, hypertriglyceridemia) and bone disease.
 - **Tertiary prevention.** Multidisciplinary surveillance to prevent complications (orthopedic, metabolic, dental, cardiac).
 - **Iatrogenic avoidance.** Caution with ZMPSTE24-inhibiting drugs (HIV protease inhibitors).
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14. Other Species / Natural Disease

- **Orthology.** *ZMPSTE24* is evolutionarily conserved (yeast Ste24 ortholog); mouse *Zmpste24* is the functional ortholog used to model the disease. The prelamin A processing pathway is conserved across mammals.
 - **Natural disease in other species.** No well-characterized naturally occurring animal analogue of MAD-B is documented; disease knowledge comes from engineered mouse models rather than spontaneous veterinary cases.
 - **Comparative biology.** *Zmpste24*^{-/-} mice recapitulate core human features (progeroid phenotype, osteoporosis, lipodystrophy, muscle wasting), supporting strong cross-species mechanism conservation.
 - **Zoonotic potential.** Not applicable (non-infectious genetic disease).
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15. Model Organisms

The principal model is the ***Zmpste24*-deficient (knockout) mouse**, which accumulates prelamin A and develops progeroid features closely paralleling human prelamin A disorders (Finding F008).

Model	Type	Key phenotype / finding	Source
<i>Zmpste24</i> ^{-/-} mouse	Global knockout	Prelamin A accumulation; osteoporosis/fractures, muscle wasting, lipodystrophy, short lifespan	multiple
<i>Suv39h1</i> -depleted <i>Zmpste24</i> ^{-/-}	Compound genetic	Improved DNA repair; ~60% lifespan extension	PMID: 23695662
<i>Rce1</i> -knockout <i>Zmpste24</i> ^{-/-}	Compound genetic	Delayed senescence; doubled median survival	PMID: 32910507
RANKL-targeted progeroid mice	Therapeutic	Prevented bone loss, improved muscle function, extended lifespan	PMID: 42572248
Prx1-Cre; <i>Zmpste24</i> conditional	Conditional KO	SSPC depletion drives bone loss (scRNA-seq)	PMID: 37407584
<i>Zmpste24</i> ^{-/-} B-lymphopoiesis	Global KO	Prelamin A impairs NF-κB-dependent B-lymphopoiesis via marrow niche	PMID: 24764515
MAD-B patient fibroblasts	In vitro	Abnormal nuclear morphology; rescued by lonafarnib	PMID: 38050983

"loss of Suv39h1 in Zmpste24(-/-) mice delays body weight loss, increases bone mineral density and extends lifespan by ~60%" — [M] [PMID: 23695662](#)

"Lack of Zmpste24, a metalloproteinase responsible for prelamin A processing, leads to progeroid features resembling HGPS." — [M] [PMID: 24764515](#)

"an antiresorptive strategy, based on RANKL targeting, ameliorated the bone loss phenotype of progeroid mice" — [M] [PMID: 42572248](#)

Recapitulation / limitations. The mouse recapitulates bone, muscle, lipodystrophy, immune, and epigenetic phenotypes and supports therapeutic proof-of-concept. Limitations include species differences in lifespan and that complete-null mouse models represent the more severe end of the ZMPSTE24-deficiency spectrum (closer to RD) rather than hypomorphic MAD-B; patient fibroblasts better capture the residual-activity MAD-B state.

Key Findings (with evidence)

1. **MAD-B is an autosomal recessive laminopathy caused by biallelic *ZMPSTE24* mutations reducing prelamin A processing (F001).** Case series and reviews consistently attribute MAD-B to compound heterozygous/homozygous *ZMPSTE24* mutations reducing zinc-metalloprotease activity and causing accumulation of unprocessed farnesylated prelamin A; the Suriname founder cohort (n=8) all carried homozygous c.1196A>G p.(Tyr399Cys). [PMID: 31856865, 21267004]
2. **Residual *ZMPSTE24* activity sets an allelic severity gradient from lethal RD to MAD-B (F002).** Complete LoF (biallelic null) → neonatally lethal restrictive dermopathy; residual-activity variants → survivable MAD-B; severity correlates with residual activity and prelamin A burden. [PMID: 41877632, 21267004]
3. **Clinical diagnostic criteria defined from a 20-patient aggregate (F003).** Major criteria present in 85–100%: short stature, clavicular hypoplasia, delayed suture closure, high palate, mandibular hypoplasia, dental crowding, acro-osteolysis, hypoplastic nails, brittle/sparse hair, mottled pigmentation, plus generalized lipodystrophy and metabolic complications. [PMID: 31856865]
4. **Prelamin A accumulation drives downstream AKT–mTOR activation, senescence, DNA-repair defects, and oxidative/mitochondrial stress (F004).** [PMID: 23686339, 21746928, 21828285]
5. **Lonafarnib (FTI) rescues nuclear morphology in MAD-B fibroblasts, supporting prenylation-targeted therapy; HGPS triple-therapy trial improved bone density and weight (F005).** [PMID: 38050983, 27400896]
6. **MAD-B sits within a gene-defined classification of mandibuloacral dysplasias/progeroid differentials (F006).** [PMID: 29208544, 32917887]

7. **ZMPSTE24 variant spectrum spans missense, frameshift, and splice variants, with N-terminal frameshifts rescued by alternative translation initiation (F007).** [PMID: 37270786, 35597529, 30919593]
 8. **The *Zmpste24*^{-/-} mouse is the principal model**, recapitulating bone, muscle, lipodystrophy, immune, and epigenetic defects and supporting therapeutic targets (F008). [PMID: 23695662, 24764515, 42572248, 37407584]
 9. **Pharmacologic ZMPSTE24 inhibition (HIV protease inhibitors) phenocopies prelamina A disease (F009).** [PMID: 40461512]
 10. **Management is multidisciplinary/symptomatic; metreleptin addresses the leptin-deficient generalized lipodystrophy and metabolic complications (F010).** [PMID: 42507316, 31856865]
 11. **MAD-B has postnatal/childhood onset with a chronic, slowly progressive course and survival into adulthood**, unlike lethal neonatal RD (F011). [PMID: 41877632, 21267004]
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Evidence Base (selected literature)

PMID	Role	Contribution
31856865	Anchor clinical	Suriname cohort, founder mutation, diagnostic criteria & management
21267004	Clinical/ mechanism	MAD-B with congenital myopathy; genotype–severity correlation; 30-yr survey
41877632	Allelic contrast	RD vs MAD-B activity gradient; lethal neonatal course of RD
37270786 / 35597529	Variant biology	Alternative translation initiation rescue converting null→hypomorph
30919593	Variant	Compound het variants (Chile)
29341437	Variant/ mechanism	Mosaic UPD chr1; phenotypic heterogeneity; intermediate form
23686339	Mechanism	AKT–mTOR activation by prelamin A; ICMT targeting
21746928	Mechanism	H4K16 hypoacetylation and DNA-repair defect
21828285	Mechanism	Adipose mitochondrial dysfunction/oxidative stress
38050983	Therapy	Lonafarnib rescues nuclear morphology in MAD-B fibroblasts
27400896	Therapy	HGPS triple-therapy trial outcomes
23695662 / 32910507 / 42572248 / 24764515 / 37407584	Models	Zmpste24 ^{-/-} mouse mechanisms & therapeutic modifiers
40461512	Environmental	HIV protease inhibitor ZMPSTE24 inhibition phenocopy
42507316	Therapy	Metreleptin metabolic benefit in generalized lipodystrophy
29208544 / 32917887	Classification	MAD A/B gene definitions; MTX2/MDPS differential

Evidence source types. Human clinical (case series/reports, diagnostic-criteria aggregate), in vitro (patient fibroblasts, FTI rescue), and model organism (mouse genetic/therapeutic studies) all contribute. Mechanistic depth is strongest from mouse models and shared prelamins A/HGPS biology; MAD-B-specific clinical data are limited by rarity.

Limitations and Knowledge Gaps

- **Small clinical denominator.** The strongest clinical dataset is ~20 genetically confirmed patients; frequencies, QoL, and natural-history metrics carry wide uncertainty. No formal prevalence/incidence exists.
 - **Mechanism extrapolation.** Much mechanistic detail (AKT–mTOR, H4K16, mitochondrial, immune) derives from *Zmpste24*^{-/-} null mice and HGPS, which model the severe end; the **hypomorphic residual-activity state specific to MAD-B** is less directly studied in vivo.
 - **No MAD-B-specific therapeutic trials.** Metreleptin and FTI evidence is extrapolated from generalized lipodystrophy and HGPS cohorts, not MAD-B trials.
 - **Genotype–phenotype quantitation.** The activity gradient is well-supported qualitatively but lacks systematic quantitative mapping of residual activity to specific phenotype severity across the full variant spectrum.
 - **Cardiac risk** in MAD-B specifically (vs HGPS/model repolarization data) is under-characterized.
 - **Validated QoL instruments** for MAD-B are absent.
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Proposed Follow-up Experiments / Actions

1. **Patient-derived iPSC and organoid models** (bone, adipose) carrying defined MAD-B hypomorphic alleles (e.g., p.Tyr399Cys) to study the residual-activity state and test prenylation-targeted drugs in a disease-specific context.
2. **Quantitative residual-activity assays** across the reported ZMPSTE24 variant spectrum, correlated to clinical severity scores, to formalize genotype–phenotype prediction.
3. **Prospective registry / natural-history study** (leveraging founder populations such as Suriname) to establish prevalence, progression rates, metabolic outcomes, cardiac surveillance needs, and validated QoL metrics.

4. **Pilot trials of metreleptin** specifically in MAD-B generalized lipodystrophy, with metabolic endpoints (HbA1c, triglycerides, hepatic fat).
 5. **Mechanism-targeted combination testing** (FTI $\pm$ statin $\pm$ bisphosphonate; RANKL inhibition; mTOR/senolytic approaches) in MAD-B models, prioritizing early-window intervention informed by mouse survival data.
 6. **Cardiac phenotyping** (ECG/repolarization, imaging) in MAD-B patients given prelamin A cardiac electrical defects seen in progeroid models.
 7. **Pharmacovigilance guidance** formalizing avoidance of ZMPSTE24-inhibiting drugs (HIV protease inhibitors) in patients with prelamin A disorders.
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Report compiled from 11 confirmed findings and 47 reviewed papers across 5 investigation iterations. Evidence spans human clinical case series, in vitro patient-fibroblast studies, and mouse model work; claims are cited to primary literature by PMID.
